Improved Design of CMOS 1-Bit Comparator With Stacking Technique

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***Abstract*—This paper presents the design and analysis of a CMOS based efficient one bit comparator circuit. Comparator has been designed and simulated in 180nm TSMC technology. The proposed circuit uses a XNOR gate, two AND gates and two NOT gates. Stacking technique has been used to improve the conventional design. Power supply has been varied from 1.0V to 2.8V for the proposed and basic design. Result reflects that proposed approach shows less power consumption and improved power delay product (PDP).**

Keywords*— CMOS, Comparator, power dissipation, Stacking, VLSI.

I. INTRODUCTION

The comparator is an electronic circuit which compares the magnitude of one signal to another signal. The comparators are used in analog to digital converters [1], [2]. Comparator is the "Heart" of the Analog to digital converter (ADC). As we talk about the VLSI technology, in this environment the device should be of higher speed, less cost and low power consumption. Many of the electronic devices like mobile devices and other portable computing devices have constraints in terms of power consumption [3]. Power consumption is one of the important factors of VLSI circuit design for CMOS is the primary technology. The power consumption has become a fundamental problem in VLSI circuit design. Therefore, reducing the power consumption of integrated circuits through design improvement is a major challenge in portable system design. To solve the power consumption problem, many different techniques from circuit level to device level and above have been proposed by researchers. However, there is no straight forward ways to meet the tradeoff between power, delay and area. In CMOS comparator design there are some points of consideration like chip size, speed, power consumption are taken into account [4]. To overcome the constraints, several logic styles have been developed to improve power consumption. Different techniques provide an efficient way to reduce leakage power, but disadvantages of each technique limit the application of each technique.

II. BASIC COMPARATOR

The comparator is basically 1-bit analog to digital converter [5]. Fig.1 shows general block diagram of comparator and Fig.2 shows symbol of comparator.

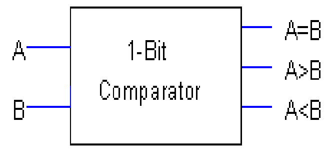

Fig.1: Block diagram of Comparator

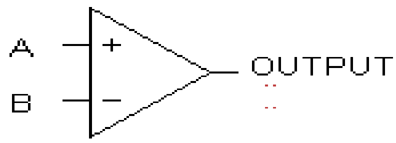

Fig.2: Symbol of Comparator

TABLE 1 TRUTH TABLE OF 1-BIT COMPARATOR

INPUT		OUTPUT		
A	B	A=B	A>B	A<B
0	0	1	0	0
0	1	0	0	1
1	0	0	1	0
1	1	1	0	0

The equation for A=B is $\mathbf{A \Theta B}$ **(1)**

The equation for A>B is **A** **(2)**

The equation for A<B is **B** **(3)**

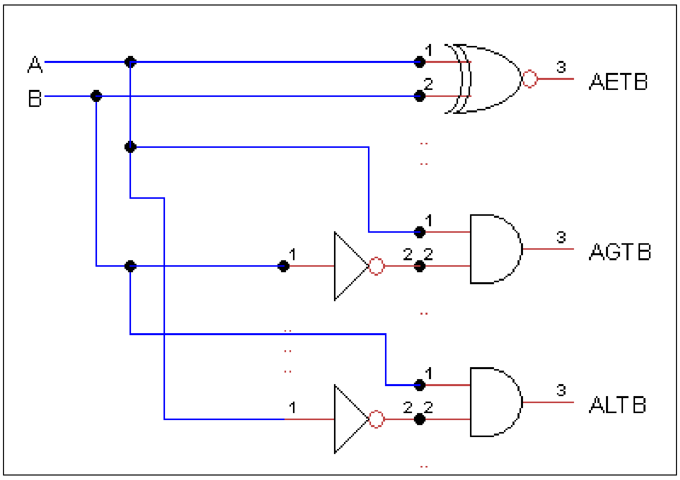

Fig.3: Logic diagram of 1-Bit Comparator

III. EARLIER WORK

The basic 1 bit Comparator circuit consist of three basic building blocks, two AND gates and one XNOR gate. Each AND gate consists of six MOSFET, three PMOS and three NMOS. XNOR gate consists of fourteen MOSFET, seven PMOS and seven NMOS.

The comparator basically a analog to digital converter, Analog signal is converted into digital signal with the help of sampling .The sampling frequency (f_s) is defined as the reciprocal of time period T.

$$\mathbf{f_s = 1/T} \qquad \mathbf{(4)}$$

The propagation delay time (t_p) of comparator is determined with the help of slew rate (SR) of the comparator.

$$tp = \frac{\Delta V}{2SR} = \frac{V_{OH} - V_{OL}}{2SR} \qquad (5)$$

where, t_p= propagation delay
SR= slew rate,
V_{OH} = upper threshold voltage
V_{OL} = lower threshold voltage

IV. PROPOSED COMPARATOR

In this work the stacking techniques has been applied to conventional comparator to reduce the power consumption. Fig.3 shows stacking technique [6], [7]. In stacking technique every transistor is divided into two transistors each have W/L is half of parent transistor as shown in the Fig.4.
The number of transistor increases but area is not affected much since W of parent transistor is divide into two transistor of W/2 width each.

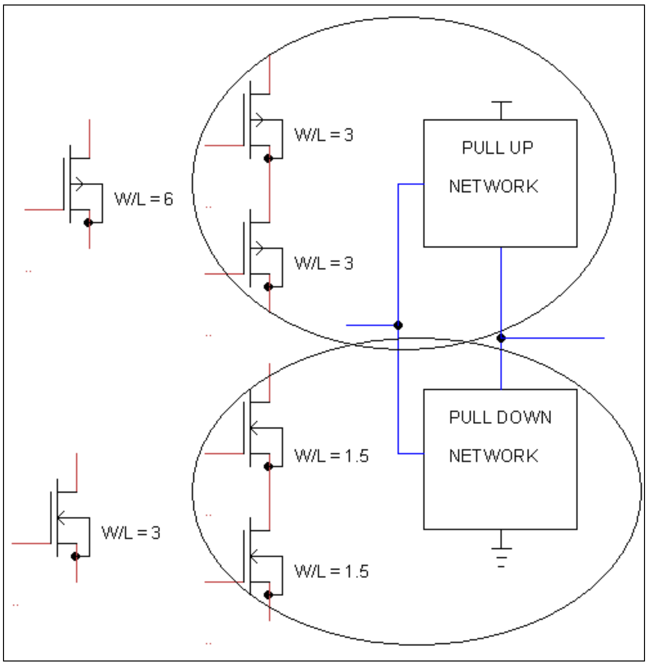

Fig.4: Stacking technique

When two transistors are turned off together induced reverse bias between the two transistors results decrease in power. This technique is based on the fact that natural stacking of MOSFET helps in achieving leakage current. The leakage through two series OFF transistor is much lower than that of single transistor because of stack effect [8]. An effective way to reduce leakage power in active mode is stacking of transistor [8]. In proposed comparator circuit there are three building blocks, two AND gates and one XNOR gate. Each AND gate consists ten MOSFETS, five PMOS and five NMOS each. XNOR gate consist of twenty two MOSFETS including eleven NMOS and eleven PMOS. The proposed comparator circuit is shown in Fig.6. For AGTB and ALTB, AND gate has been used because in AND gate whenever both inputs are high then output will be high otherwise output will be zero. For AETB a XNOR gate has been used as in XNOR gate whenever both inputs are same then output will be high otherwise it will be zero. The basic comparator circuit shown in Fig.5 and proposed comparator circuit shown in Fig.6 In n Bit comparator the truth table consists $\mathbf{2^{2n}}$ terms in which number of terms for equal condition will be $\mathbf{2^n}$ and for greater than and less than number of terms will be

$$\frac{2^{2n} - 2^n}{2} \qquad (6)$$

In 1 bit comparator number of AETB conditions are 2 and number of AGTB or ALTB conditions are 1 as shown in TABLE 1.

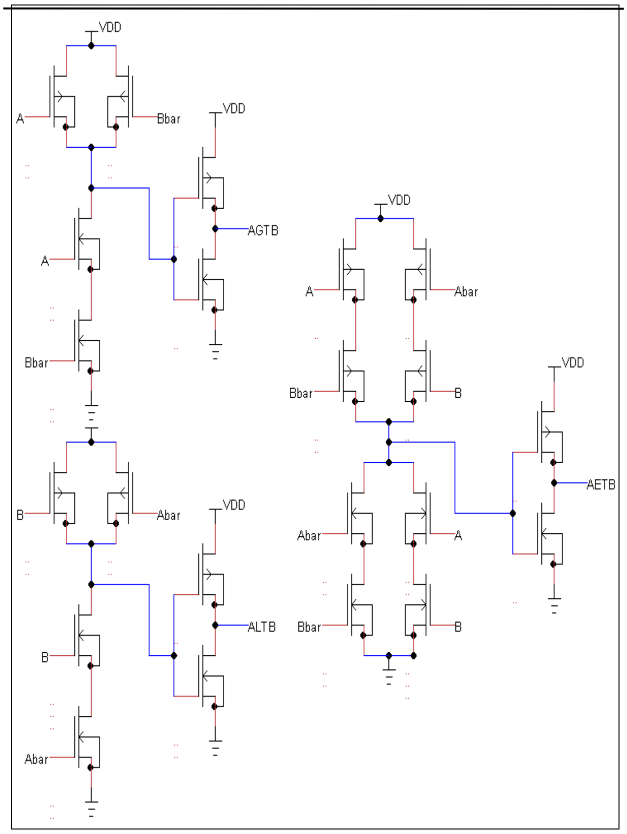

Fig.5: Basic comparator circuit

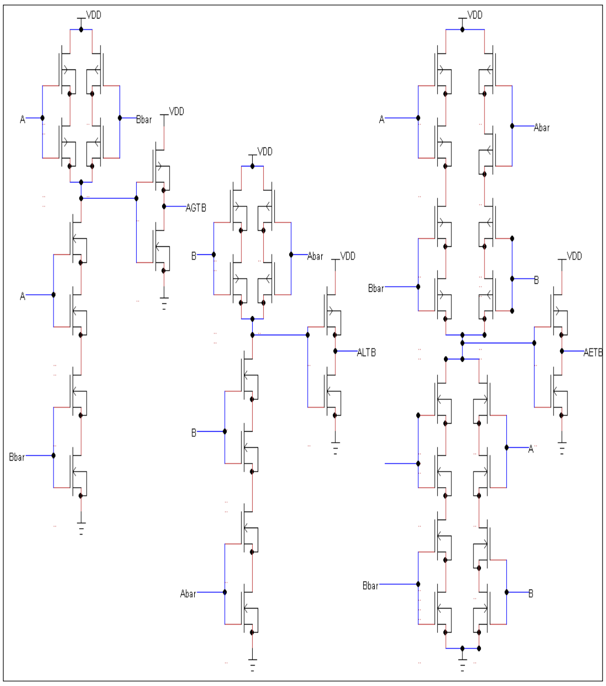

Fig.6: Proposed comparator circuit

V. RESULTS AND DISCUSSIONS

Results of proposed and conventional comparator has been obtained in TSMC 180nm technology for different power supply voltages varying from 1.0 V to 2.8V. The simulation waveform of 1-bit conventional comparator is shown in Fig.7.

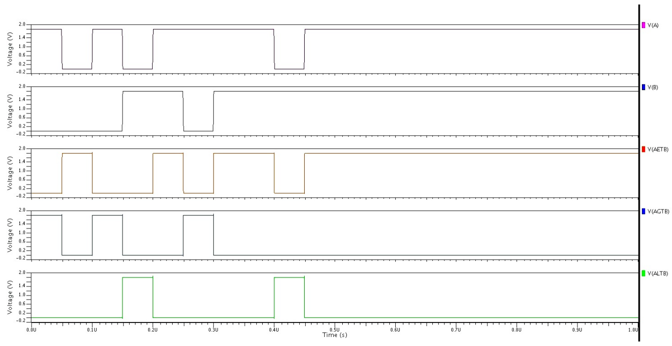

Fig.7: Waveform results of 1-bit conventional comparator

The simulation waveform of 1-bit proposed comparator is shown in Fig.8.

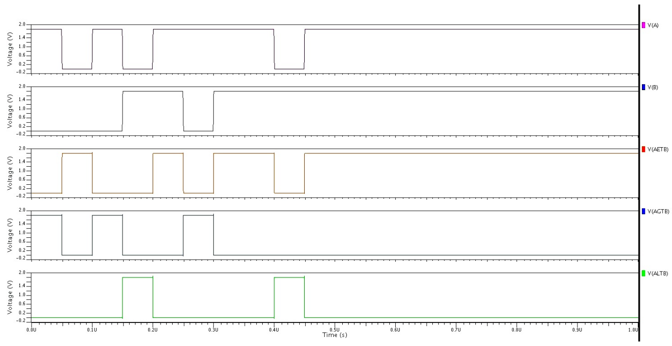

Fig.8: Schematic simulation of proposed 1-Bit Comparator

The comparative analysis of power consumption between two comparators is shown in TABLE 2. It has been observed that as V_{dd} increases the power consumption increases because power is directly proportional to V_{dd}^2.

TABLE 2 COMPARISSON OF POWER CONSUMPTION

Vdd (V)	Basic Comparator(pW)	Proposed Comparator(pW)
1.0	93.06	54.58
1.2	129.84	76.9
1.4	174.05	103.92
1.6	226.45	136.21
1.8	287.93	174.35
2.0	359.46	219
2.2	442.1	270.93
2.4	537.04	331
2.6	645.56	400.16
2.8	769.1	479.55

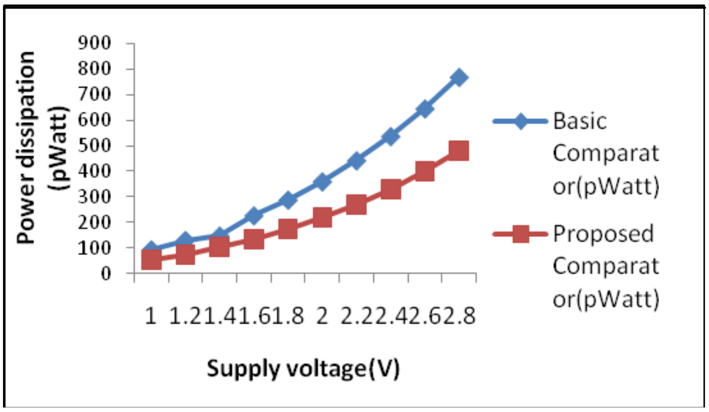

Fig 9: Variation of Power dissipation in Basic Comparator and Proposed Comparator.

The comparative analysis in terms of delay between two comparators is shown in TABLE 3.It has been observed that as V_{dd} increases delay is increases.

TABLE 3 COMPARISION OF DELAY

V_{dd} (V)	Basic Comparator (nS)	Proposed comparator (nS)
1.0	49.21	48.54
1.2	49.45	48.98
1.4	49.56	49.19
1.6	49.63	49.34
1.8	49.67	49.38
2.0	49.7	49.47
2.2	49.71	49.49
2.4	49.73	49.52
2.6	49.74	49.54
2.8	49.75	49.56

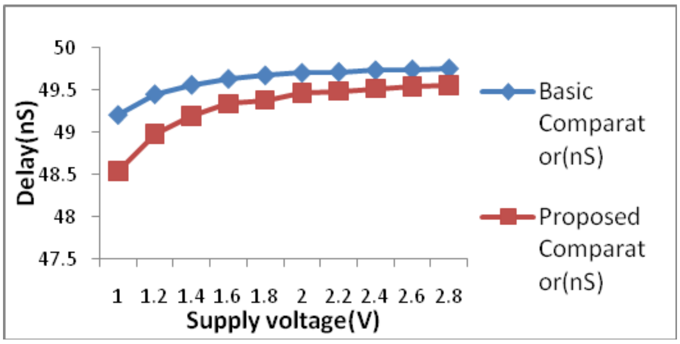

Fig.10: Variation of delay in basic comparator and proposed Comparator

TABLE 4 shows comparison in terms of power, delay and power delay product. Power consumption in basic Comparator circuit is 287.93pWatt whereas the proposed Comparator consumes only 174.352pW this shows their significant power saving in proposed design. It is also observe that the delay in case of basic comparator is 49.67ns where as the delay in proposed comparator is 49.38ns there is very small improvement in the delay of proposed circuit. Further, the PDP also shows improvement in the proposed design.

TABLE 4 COMPARISION OF DIFFERENT COMPARATORS

Approach type	Power(pWatt)	Delay(nS)	PDP(atto Joule)
Basic Comparator	287.93	49.67	14.3
Proposed Comparator	174.352	49.38	8.6

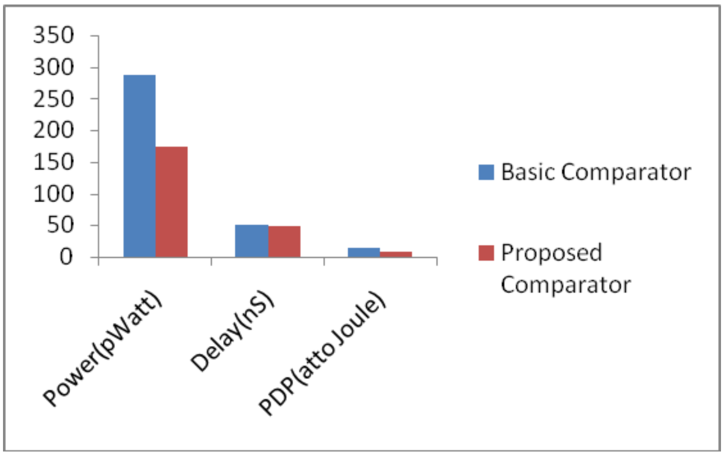

Fig.11: Comparison of basic and proposed approach

VI. CONCLUSION

In this work a new design of comparator has been proposed and power consumption, delay and PDP results have been obtained for the basic design and proposed circuit. The Proposed Comparator has less power consumption, less delay and less PDP as compared to basic comparator. It is noticed that the power consumption in proposed comparator is 174.352pWatt whereas the power consumption in basic comparator circuit is 287.93 pW, in terms of delay the delay in both circuits are almost same there is very little difference, PDP of proposed comparator is smaller than basic comparator. Proposed comparator shows less power consumption and improved power delay product.

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